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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Clinic Prescription Record and Patient Visit Tracking System

Submitted by

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**CLINIC PRESCRIPTION RECORD AND PATIENT VISIT TRACKING SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java programming language** is a **record of my original work** carried out under the guidance of **Dr.Boopalan Sundaramoorthy,** Technical Head, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Clinic Prescription Record And Patient Visit Tracking System"** is a **bona fide work** carried out by **Lydia Janet.C**, I B.sc CS **[2025 - 2028]**, **Department Of Computer Science, KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming Language**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMABATORE

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on **10/04/2026**

Guide:
(Technical Head, KGCAS)

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Head of the Department
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ABSTRACT

The **Clinic Prescription Record and Patient Visit Tracking System** is a Java-based application developed using Java Swing to provide a user-friendly graphical interface for managing patient records and clinic operations. In many clinics, patient registration and prescription handling is handled manually, which can result in data duplication, errors, and difficulty in searching and managing records. This project aims to automate and simplify the entire patient registration and visit tracking process.

The system allows users to register **patients** by entering details such as **patient ID, name, age, disease, prescription, branch, and time slot**. It includes features like adding new **patients**, **preventing duplicate time slot bookings, managing patient queue, and notifying the next patient**, and searching patient records using partial name or ID matching. The application also provides a dashboard that displays the total number of patients waiting, ensuring efficient clinic workflow management.

The project is implemented using Core Java programming concepts such as classes, objects, **ArrayList collections and HashSet** (for slot management), file handling (serialization), exception handling, and event-driven programming with Java Swing components like JTable, JTextField, and JButton. The system ensures data persistence by storing patient details in a file, allowing data to be saved and loaded efficiently.

The main objective of this system is to improve accuracy in patient data management, reduce manual effort, and provide quick and efficient access to patient and prescription information. It enhances user experience through features like organized patient scheduling and digital prescription handling.

This project demonstrates the practical application of **object-oriented programming concepts and GUI development in Java** and serves as a foundation for building more advanced **healthcare management systems** with additional features like PDF prescription generation, database integration, and real-time analytics.

1. INTRODUCTION

In the healthcare sector, efficient patient record management systems play a crucial role in ensuring smooth clinic operations and quality patient care. Traditionally, many small clinics rely on manual methods for recording patient details, prescriptions, and visit schedules, where staff maintain records using paper files or basic registers. This manual process is time-consuming and can often lead to data duplication, errors, difficulty in retrieving patient history, and poor management of appointment schedules.

With the advancement of computer technology, automated patient management systems have become essential for improving efficiency and accuracy in clinic operations. A computerized system helps clinic staff quickly register patients, manage appointments, generate prescriptions, and maintain organized digital records of patient visits and treatments.

The **Clinic Prescription Record and Patient Visit Tracking System** is a Java-based application designed to automate the process of patient registration, prescription management, and visit tracking. The system allows the user to enter patient details such as ID, name, age, disease, prescription, branch, and time slot. It also ensures efficient appointment scheduling by preventing duplicate time slot bookings and managing patients using a queue-based system. The application provides a clear and structured interface to view patient records and track ongoing consultations.

This project is developed using **Core Java programming concepts** such as classes, objects, collections, file handling (serialization), conditional statements, exception handling, and event-driven programming with Java Swing for building a **graphical user interface (GUI)**.

The primary goal of this system is to reduce manual effort, improve accuracy in patient data management, and enhance the overall efficiency of clinic operations. This project also demonstrates how Java programming can be applied to develop practical real-world healthcare applications.

1.1 Objective

The main objective of the Clinic Prescription Record and Patient Visit Tracking System is to develop a simple and efficient application using **core Java** that automates the process of **patient registration, prescription management,** and **visit tracking** in clinics.

The specific objectives of this project are:

- To design and develop a **Java-based clinic management system**.
- To automate **patient registration** and maintain **digital records.**
- To manage **appointments** and prevent **duplicate bookings.**
- To implement a queue system to track and **notify patients.**
- To generate **digital prescriptions.**
- To reduce manual errors in **data entry** and **scheduling.**

- To provide a **user-friendly interface** using **Java Swing.**
- To implement **Core Java concepts** like **collections** and **file handling**.
- To store and retrieve **patient data efficiently.**
- To improve overall **clinic workflow** and **efficiency**.

1.2 Overview

The **Clinic Prescription Record and Patient Visit Tracking System** is a **menu-driven application** developed using the **Java programming language.** The system provides a **digital platform** where **clinic staff** can **register patients**, **manage appointments**, and maintain **prescription records** efficiently.

The system allows the user to enter **patient details** such as **ID, name, age, disease, prescription, branch,** and **time slot**. It ensures proper appointment scheduling by assigning time slots and preventing duplicate bookings. The application also maintains a **queue system** to track patients and **notify the next patient** for consultation.

The system helps simplify clinic operations by eliminating **manual record keeping** and reducing the chances of errors in patient data and scheduling. The application is developed using **Java Swing** to provide a **graphical user interface (GUI)** for easy interaction.

This project demonstrates the practical use of **Java programming** in developing **real-world healthcare applications** for clinics and small medical centers.

1.3 Features

The **Clinic Prescription Record and Patient Visit Tracking System** includes the following main features:

• **Patient Registration** – Allows the user to enter and store patient details such as ID, name, age, disease, and prescription.

• **Time Slot Management** – Enables appointment scheduling by assigning time slots and preventing duplicate bookings.

• **Queue Management** – Maintains patients in a queue and notifies the next patient for consultation.

• **Prescription Generation** – Generates and stores digital prescription details for each patient.

• **Patient Record Display** – Displays all patient details in a tabular format using a GUI table.

• **Dashboard Display** – Shows the total number of patients currently waiting.

• **Data Storage** – Saves and loads patient records using file handling (serialization).

• **User-Friendly Interface** – Provides an easy-to-use graphical interface using Java Swing components.

• **Error Handling** – Validates input data and prevents incorrect entries.

• **Efficient Workflow** – Improves clinic operations by reducing manual work and saving time.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the system requirements are identified and understood. It helps in defining how the system should work effectively.

In clinics, manual record keeping leads to errors, difficulty in managing patient data, and appointment issues. This system provides a computerized solution to automate patient registration, prescription management, and visit tracking, improving accuracy and efficiency.

2.1 Problem Definition

In many small clinics, patient record management and appointment handling are often performed manually using paper records or basic registers. This traditional method creates several challenges.

Some common problems include:

• Errors in maintaining patient records
• Time-consuming registration process
• Difficulty in managing appointments and time slots
• Lack of proper prescription records
• Inefficient patient handling during busy hours

Manual systems also increase the workload for clinic staff and can lead to delays and confusion in patient management.

Therefore, there is a need for an automated system that can efficiently manage patient records, schedule appointments, and maintain prescriptions.

The **Clinic Prescription Record and Patient Visit Tracking System** is designed to solve these problems by providing a computerized solution that automates patient registration, prescription management, and visit tracking.

2.2 Project Overview

The **Clinic Prescription Record and Patient Visit Tracking System** is a Java-based application developed to simplify patient management in clinics.

The system provides a user-friendly interface where users can enter patient details and manage appointments efficiently. It allows clinic staff to assign time slots, maintain patient records, and generate prescription details. The system also manages patient flow using a queue-based approach and displays patient information clearly.

The system includes the following functionalities:

• Registering patient details
• Assigning time slots and preventing duplicate bookings
• Managing patient queue and notifying the next patient
• Generating and storing prescription details
• Displaying patient records in a structured format
• Saving and loading patient data using file handling

This project is implemented using **Core Java programming** concepts and is developed as a **Java Swing GUI application.**

2.3 Scopes and Limitations

Scope

The scope of the **Clinic Prescription Record and Patient Visit Tracking System** includes the following:

• Managing patient registration details
• Storing and organizing patient records efficiently
• Scheduling appointments with time slot management
• Preventing duplicate time slot bookings
• Managing patient queue and visit tracking
• Generating and maintaining digital prescription records
• Displaying patient details through a dashboard
• Demonstrating the practical use of Java programming concepts

The system can be used in:

• Small clinics
• Private hospitals
• Medical centers
• Diagnostic centers
• Outpatient departments

Limitations:

• The system uses file handling instead of a database, which is not suitable for very large data
• No online appointment booking or web-based access
• Limited to single-user operation
• No advanced security or authentication features
• Requires manual data entry

3. SYSTEM DESIGN

System design is the phase where the **structure** and **workflow** of the system are planned before implementation. It helps in organizing the system into different modules for efficient development.

The **Clinic Prescription Record and Patient Visit Tracking System** is designed as a **GUI-based application** to manage patient records, appointments, and prescriptions.

The system is divided into **modules** such as patient registration, time slot management, queue handling, and data storage.

3.1 Modules

The main modules of the **Clinic Prescription Record and Patient Visit Tracking System** are explained below:

3.1. Patient Registration Module

This module is responsible for entering and storing patient details such as ID, name, age, disease, prescription, branch, and time slot.

Functions of this module include:
• Accepting patient details
• Validating input (ID and age must be numeric)
• Storing patient data in the system

The data is saved so that it can be used later for management and retrieval.

3.2. Display Module

This module displays all registered patient records in a table format.

Functions of this module include:
• Showing all patient details
• Organizing data in rows and columns
• Providing a clear view of stored records

The table updates automatically after every operation.

3.3. Time Slot Management Module

This module manages appointment scheduling using time slots.

Functions of this module include:
• Assigning time slots to patients
• Preventing duplicate bookings using slot checking
• Managing appointments efficiently

This ensures proper scheduling without conflicts.

3.4. Queue Management Module

This module handles patient flow based on queue order.

Functions of this module include:
• Maintaining patients in first-come-first-serve order
• Notifying the next patient for consultation
• Removing served patients from the queue

This helps in smooth patient handling.

3.5. Prescription Module

This module generates and stores prescription details.

Functions of this module include:
• Creating digital prescription records
• Storing prescription information with patient data
• Displaying prescription details

This ensures proper medical record keeping.

3.6. Dashboard Module

This module provides an overview of patient count.

Functions of this module include:
• Displaying total number of patients waiting

• Updating count dynamically
• Providing quick summary information

This helps in monitoring clinic workload.

3.7. Data Storage Module

This module handles saving and loading of patient data.

Functions of this module include:
• Saving patient records using file handling (serialization)
• Loading saved data when the system starts
• Maintaining data persistence

This ensures data is not lost after closing the application.

3.8. Exit Module

This module allows the user to safely close the application.

When the user exits the system, the program terminates after saving the data.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing,implementing,testing, and documenting the system.

For the **Clinic Prescription Record and Patient Visit Tracking System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of clinic operations.

The main problems identified in manual systems are:

• Errors in maintaining patient records
• Time-consuming patient registration process
• Difficulty in managing appointments and time slots
• Lack of proper prescription records

Based on these problems, the system requirements were defined to develop an automated application that simplifies patient management and visit tracking.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins. The system is divided into different modules to handle specific functionalities efficiently.

The system is divided into modules such as:

• Patient Registration Module
• Queue Management Module
• Appointment / Time Slot Module
• Prescription Module
• Data Storage Module
• Dashboard Module
• Exit Module

The program flow and structure are designed in such a way that clinic staff can easily interact with the system for managing patient records, appointments, and prescriptions.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of GUI using Java Swing
• Writing code for patient registration and slot management
• Handling user input and validation
• Displaying patient records and dashboard

Core Java concepts such as classes, objects, collections (ArrayList, HashSet), file handling, and event handling were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system stores patient details, manages appointments, and generates prescription records efficiently.

The results are demonstrated through **program execution** and **screenshots** showing the input and output of the system.

Proper documentation is prepared to explain the **design, implementation,** and **working** of the system.

4.6. Algorithm

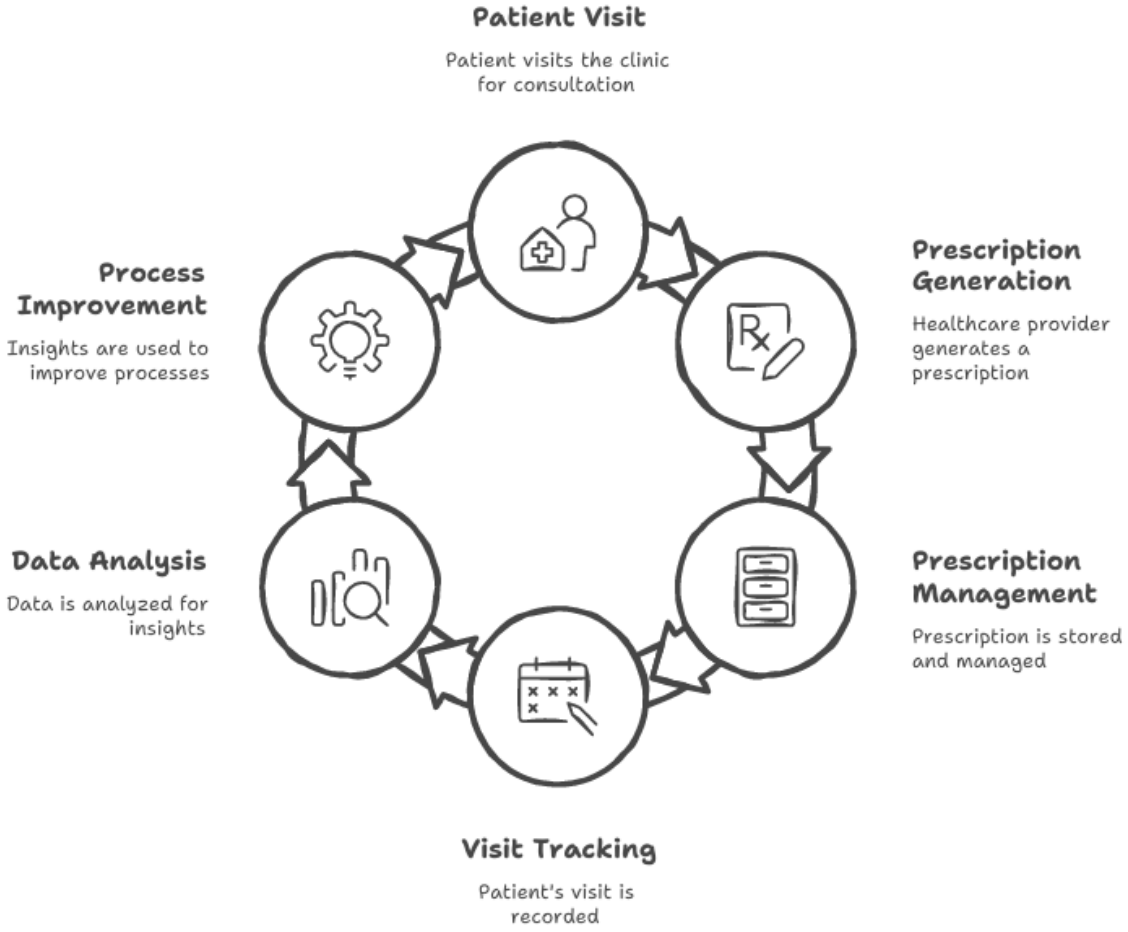
Algorithm for the Clinic Prescription Record and Patient Visit Tracking System:

1. Start the program.

2. Display the main interface with options: Add Patient, Save, Notify Next, Dashboard.
3. If the user selects **Add Patient**, accept details: ID, Name, Age, Disease, Prescription, Branch, Time Slot.
4. Validate the **input** (ID and Age must be numeric).
5. Check if the **selected time slot** is already booked.
6. If the slot is available, save **patient details** in the system and add to queue.
7. If the slot is already booked, display an error message.
8. Display all patient records in a table format.
9. If the user selects **Notify Next**, display and remove the first patient from the queue.
10. Update the dashboard with the current number of patients.
11. If the user selects **Save**, **s**tore all patient data using file handling.
12. Load saved data when the system starts.
13. Repeat steps until the user closes the application.
14. End the program.

4.7. Flowchart

Clinic Prescription Record And Patient Visit Tracking System



5. SOURCE CODE

Main Class :

```
public class ClinicGUI extends JFrame {

  public static void main(String[] args) {
     SwingUtilities.invokeLater(ClinicGUI::new);
  }
}
```

Menu Logic : (Add Patient & Slot Checking)

```
void addPatient() {

  int id = Integer.parseInt(idField.getText());
  String name = nameField.getText();

  String branch = branchBox.getSelectedItem().toString();
  String slot = timeSlotBox.getSelectedItem().toString();

  String uniqueSlot = branch + "-" + slot;

  // Prevent duplicate booking
  if (bookedSlots.contains(uniqueSlot)) {
     JOptionPane.showMessageDialog(this, "Slot already booked!");
     return;
  }
```

```
  bookedSlots.add(uniqueSlot);
}
```

Calculation Logic : (Dashboard)

```
int totalPatients = patients.size();

long chennaiCount = patients.stream()
      .filter(p -> p.branch.equals("Chennai")).count();

long coimbatoreCount = patients.stream()
      .filter(p -> p.branch.equals("Coimbatore")).count();

long doctorKumarCount = patients.stream()
      .filter(p -> p.doctor.equals("Dr. Kumar")).count();

long doctorRaviCount = patients.stream()
      .filter(p -> p.doctor.equals("Dr. Ravi")).count();
```

Example Format

```
totalPatients = patients.size();
chennai = totalPatients * 0.5;   // example logic
```

6. INPUT AND OUTPUT SCREENS

Figure 1: Main Menu Screen

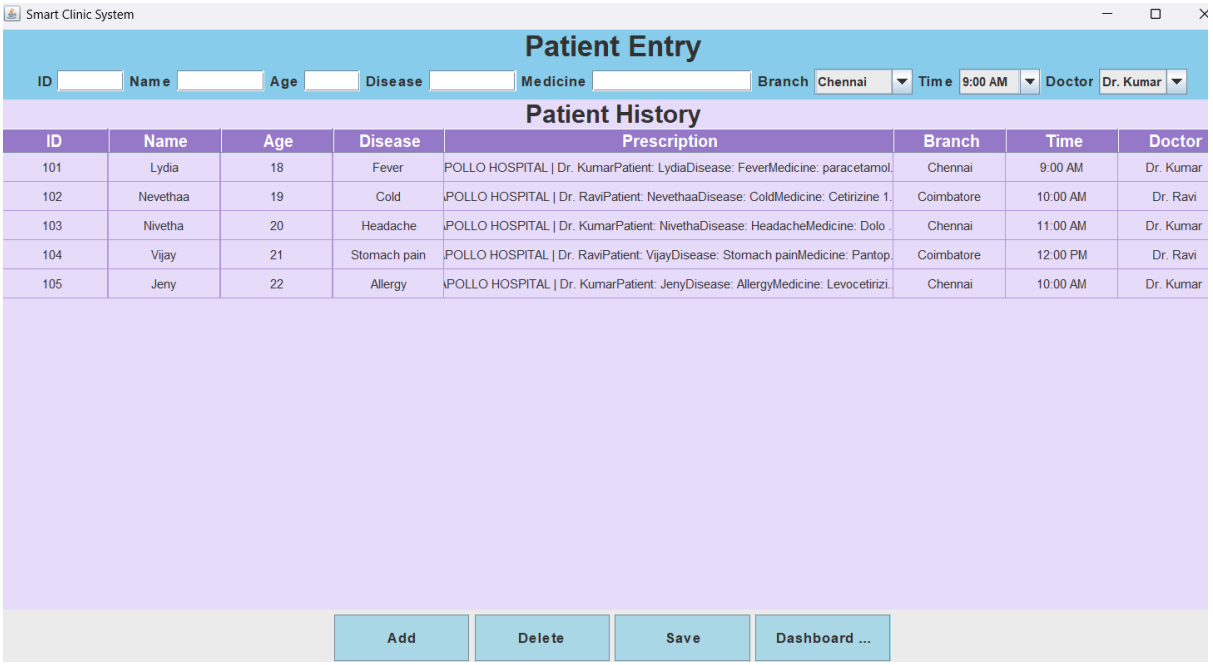


Figure 2: Patient Registration Screen

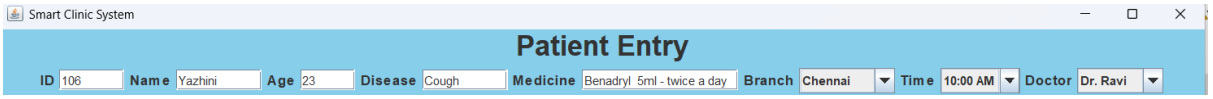


Figure 3: Display / Table Output Screen

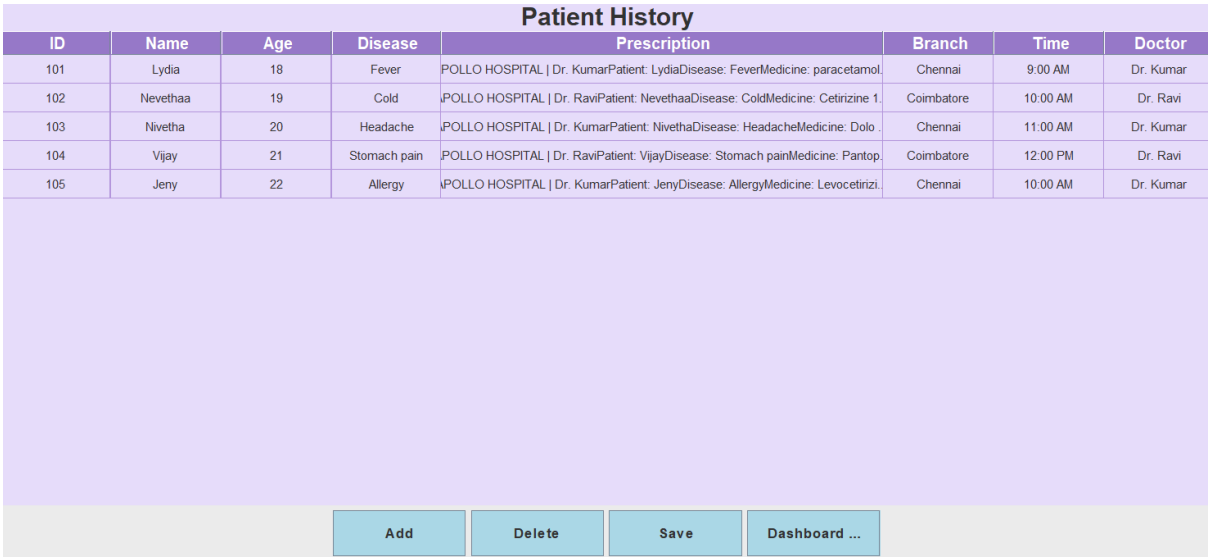
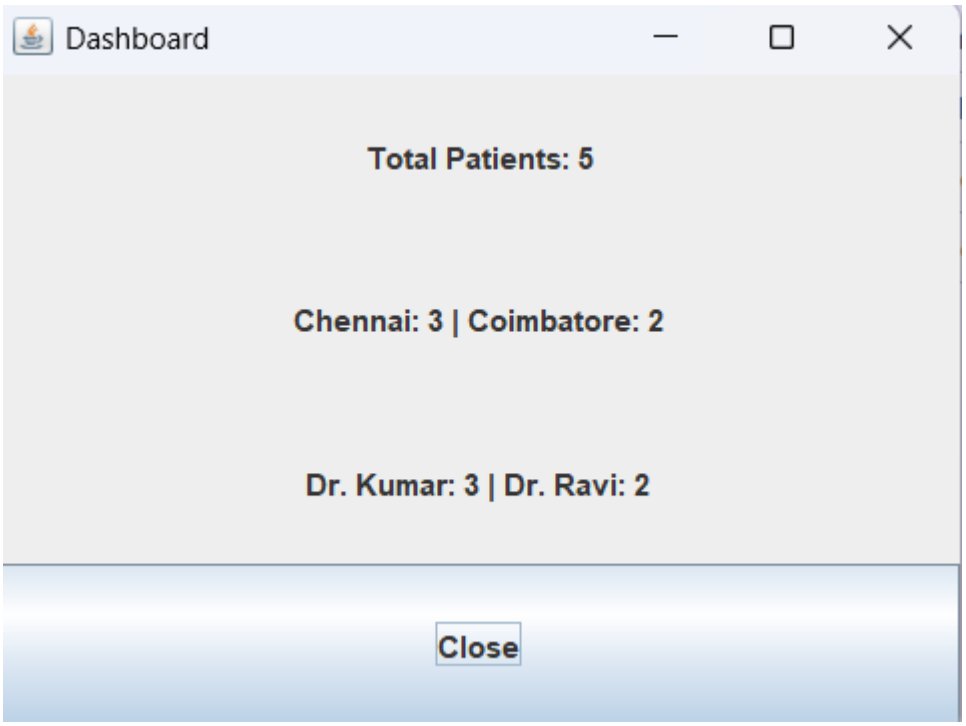


Figure 4 : Dashboard



7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Clinic Prescription Record and Patient Visit Tracking System** was developed to automate patient registration, appointment scheduling, and prescription management in clinics. The system allows users to add patient details, assign time slots, generate prescriptions, and manage patient visits efficiently.

During the development process, **Core Java concepts** such as **classes, objects, ArrayList, file handling** (serialization), and **event-driven programming** using Java Swing were implemented. The system successfully handles patient data, prevents duplicate slot booking, and maintains a queue for patient consultation.

The system was tested with different inputs to ensure proper functionality such as **patient registration, slot allocation,** and **data storag**e. The results show that the system reduces manual work, avoids errors, and improves clinic management efficiency.

7.2 Conclusion

The **Clinic Prescription Record and Patient Visit Tracking System** successfully demonstrates the practical application of Core Java in developing a real-world healthcare application.

The system automates patient management, reduces manual errors, and improves overall workflow in clinics. It provides features like patient record storage, appointment scheduling, queue management, and prescription generation.

This project helps in understanding how Java can be used to build efficient and user-friendly applications. The system can be further enhanced by adding features such as database integration, online appointment booking, and real-time notifications.

8. REFERENCES

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